Lan Peng

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RESEARCH INTERESTS

Operations research and optimization, with a focus on drone routing, coordinated vessel and drone operations, and routing with neighborhoods for maritime monitoring.

EDUCATION

- **Ph.D., Industrial Engineering**,
  University at Buffalo, SUNY, Department of Industrial and Systems Engineering, Buffalo, NY, USA (09/2018–02/2023).
- **Master's Degree, Control Science and Engineering**,
  Beihang University, School of Reliability and Systems Engineering, Beijing, China. (09/2015–03/2018).
- **Bachelor's Degree, Quality and Reliability Engineering**,
  Beihang University, School of Reliability and Systems Engineering, Beijing, China. (09/2011–07/2015).

EMPLOYMENT HISTORY

- **Assistant Professor and Master's Supervisor**,
  Shanghai University, Department of Management Science and Engineering, School of Management, Shanghai, China (04/2023–Present).
- **Teaching and Research Assistant**,
  University at Buffalo, SUNY, Department of Industrial and Systems Engineering, Buffalo, NY, USA (2018–2023).

PEER-REVIEWED PUBLICATIONS

*Corresponding author; †doctoral or master's advisor; #supervised student.

Published and Accepted

1. Weiwei Cui, Jianhui Zhu, and **Lan Peng***. "On the service time in the close-enough traveling salesman problem." *European Journal of Operational Research*, 2026, in press. [Link]
2. **Lan Peng** and Chase Murray*†. "VeRoViz: A vehicle routing visualization toolkit." *INFORMS Journal on Computing*, 34(4), 1842–1848, 2022. [Link]

Under Review

1. Weiwei Cui, Tianyi Zhang, and **Lan Peng***. "Deploying communication and relay UAVs as a flying service network: applications in emergency response."
2. **Lan Peng**, Xueqi Li#, and Lu Zhen*. "A branch-and-bound approach for the close-enough traveling salesman problem with arbitrary time-dependent neighborhoods."
3. **Lan Peng** and Lu Zhen*. "On the segment close-enough traveling salesman problem."

Working Papers

1. Weiwei Cui, Miaomiao Zou, and **Lan Peng***. "The close-enough orienteering problem with service time."
2. **Lan Peng**, Xueqi Li$^{\#}$, and Xinwei Wang*. "On the dynamic time-dependent neighborhood close-enough traveling salesman problem."
3. **Lan Peng**, Jiale Chen$^{\#}$, and Weiwei Cui*. "On the traveling bi-salesman problem."
4. Bochen Wang and **Lan Peng***. "The generalized close-enough (team) orienteering problem."

Refereed Conference Proceedings

1. **Lan Peng**, Lin Ma†, and Naichao Wang. "A fleet-level selective maintenance model for long-distance highway transportation considering stochastic repair quality." *2017 2nd International Conference on System Reliability and Safety (ICSRS)*, Milan, Italy, 2017.
2. **Lan Peng**, Baocheng Liu, Lin Ma†, Naichao Wang, and Qiannan Liu. "Mixed arithmetic reduction model for two-unit system maintenance." *2017 Second International Conference on Reliability Systems Engineering (ICRSE)*, Beijing, China, 2017.

FUNDING

Principal Investigator

1. **Young Scientists Fund (C)**, National Natural Science Foundation of China. Grant 72501172; Jan. 2026–Dec. 2028; CNY 300,000.
2. **Shanghai Magnolia Talent Program Pujiang Project**, Humanities and Social Sciences Innovation Category. Shanghai Municipal Human Resources and Social Security Bureau. Grant 24PJC034; Dec. 2024–Nov. 2027; CNY 150,000.
3. **Young Talent Start-up Program**, Shanghai University. Jan. 2024–Dec. 2026; CNY 150,000.
4. **Contract Research Funding**, Beihang University. Apr. 2025–June 2026; CNY 450,000.

GRADUATE STUDENTS

- **Master's Students, Shanghai University (In Progress)**
  - Xueqi Li (2024–Present). Drone routing for remote monitoring of moving vessels.
  - Jiale Chen (2025–Present). Drone routing in complex battlefield environments.
  - Zhipeng Zhang (2026–Present). Drone coverage routing for small maritime area targets (tentative).
  - Yuechen Lin (2026–Present). Drone coverage routing for maritime line-segment targets (tentative).
- **Master's Students, Shanghai University (Graduated)**
  - Yumeng Jiang (entered 2024). Integrated warehouse location and inventory optimization for instant retail.

TEACHING EXPERIENCE

- **Shanghai University, Course Instructor (2023–Present)**
  - Integer Programming (Winter 2023, Winter 2024, Spring 2026) [Link].
  - Operations Management (Winter 2023, Winter 2024, Winter 2025).
  - Economic Game Theory (Spring 2026).
  - E-commerce (Winter 2023).
  - Enterprise Resource Planning and Customer Relationship Management (Spring 2025).

  - Information Systems Project Management (Spring 2025, Spring 2026).
  - Management Statistics (Spring 2025, Spring 2026).
- **University at Buffalo, SUNY, Teaching Assistant (Sept. 2018–Jan. 2023)**
  - Network Optimization.
  - Facilities Planning.
  - Programming for Research.
  - Introduction to Operations Research (x 2).
  - Engineering Economy.
  - Systems Modeling and Operations Research II.
  - Behavioral Models of Social Networks.
  - Sustainable Manufacturing.
  - Introduction to Industrial Engineering.
  - Research Seminar.

CONFERENCE PRESENTATIONS

- **Lan Peng**. "A branch-and-bound approach for the close-enough traveling salesman problem with arbitrary time-dependent neighborhoods." POMS International Conference in China, Xi'an, July 2026.
- **Lan Peng**. "TDN-CETSP: A Heuristic Approach for the Time-Dependent Neighborhood Close-Enough TSP." International Conference on Optimization: Techniques and Applications, Shanghai, Apr. 2026.
- **Lan Peng** and Chase Murray. "Parallel Drone Scheduling Traveling Salesman Problem Considering Winds and Rains." INFORMS Annual Meeting, Indianapolis, IN, USA, Oct. 2022.
- **Lan Peng** and Chase Murray. "Parallel Drone Scheduling Traveling Salesman Problem with Weather Impacts." INFORMS Annual Meeting, virtual, Nov. 2021.
- **Lan Peng** and Chase Murray. "Optimization of Pick-up and Delivery Orders Bundling Problem." INFORMS Annual Meeting, virtual, Nov. 2020.
- Chase Murray and **Lan Peng**. "A Vehicle Routing Visualization Toolkit for Drones." INFORMS Annual Meeting, Seattle, WA, USA, Oct. 2019.

PROFESSIONAL SERVICE AND MEMBERSHIPS

- **Member**, Institute for Operations Research and the Management Sciences (INFORMS).
- **Member**, Systems Engineering Society of China (Committee Member, Ship and Marine Systems Engineering).